

What this result assumes

Single grid

All sixteen points ran at one mesh level, 39680 cells across 3 regions. A discretisation error bar needs at least three grids; none was run, so no numerical uncertainty is quoted for any row and none is estimated, interpolated or borrowed.

Steady points only

Each of the sixteen points is a separate converged steady state. Nothing here says how this body warms up, how long it takes to reach these temperatures, or how it responds to a change in load.

Radiation neglected

Radiation is switched off in all three regions (radiation off, radiationModel none), so heat leaves the body by conduction and forced convection alone. At the hottest surface temperature solved here a radiative path exists in the real part and is not represented.

Representative geometry and materials

The body is a representative motor-in-duct wedge with constant conductivities of 40, 167 and 0.026 W/mK for core, housing and air. It is not a drawing of any particular product and the properties are not a supplier data sheet.

Turbulence is modelled, not resolved

The air side uses the k-omega SST closure. Turbulent transport is represented by a model, so the convective heat transfer carries that model's error, and no model-form uncertainty is quoted here.

The peak is in the core

The hottest solid cell is in the core at 107.7 degC, 4.1 K above the housing peak. The margin to the 200 degC limit is computed on the core; the housing value is reported beside it and is not the hottest point.

Mesh identity is checked, not assumed

All 16 cases byte-identical in fluid, housing and core -- 48 of 48 region-case pairs.